Deep Learning & Python – Multiple Choice Assessment

30 Questions 15 Neural Networks 15 Python

Section A: Neural Networks

Questions 1–15 · 15 marks

1. What is the primary purpose of the bias term in an artificial neuron?

A. To determine the number of inputs

B. To shift the activation function and allow a non-zero output when inputs are zero

C. To calculate the learning rate

D. To normalize all input values

2. For a neuron with x_1=2, x_2=3, w_1=0.5, w_2=0.2 and bias=1, what is the output before activation?

A. 1.6

B. 2.0

C. 2.6

D. 3.5

3. In the linear regression equation y = mx + c, what does m represent?

A. The intercept

B. The slope or rate of change

C. The loss function

D. The predicted error

4. Why are activation functions used in hidden layers of neural networks?

A. To eliminate all weights

B. To introduce non-linearity

C. To increase the dataset size

D. To avoid forward propagation

5. What happens during the forward pass of a neural network?

A. Weights are updated using gradients

B. Loss is propagated backward

C. Inputs are processed through layers to generate predictions

D. Training data is shuffled only

6. What is the main purpose of backpropagation?

A. To calculate gradients of loss with respect to parameters

B. To generate random training samples

C. To remove hidden layers

D. To increase the batch size

7. A dataset has 1,000 samples and a mini-batch size of 100. How many parameter-update steps occur in one epoch?

A. 1

B. 10

C. 100

D. 1,000

8. Which statement correctly describes stochastic gradient descent (SGD) in its single-sample form?

A. It uses the entire dataset for each update

B. It updates parameters using one training sample at a time

C. It does not require gradients

D. It always converges in one epoch

9. What is the output range of the sigmoid activation function?

A. -1 to 1

B. 0 to 1

C. 0 to infinity

D. −infinity to infinity

10. Which activation function is commonly used to produce probabilities across mutually exclusive classes?

A. ReLU

B. Linear

C. Softmax

D. Leaky ReLU

11. What is the output of ReLU when the input is −4?

A. −4

B. 0

C. 0.5

D. 4

12. What is the main advantage of Leaky ReLU over standard ReLU?

A. It always outputs values between 0 and 1

B. It allows a small non-zero gradient for negative inputs

C. It removes the need for weights

D. It converts outputs into probabilities

13. What is the main difference between a derivative and a partial derivative?

A. A derivative applies only to linear functions, while a partial derivative applies to nonlinear functions.

B. A derivative measures the rate of change with respect to one variable, while a partial derivative measures the rate of change with respect to one variable while holding other variables constant.

C. A derivative is used only in forward propagation, while a partial derivative is used only in backpropagation.

D. Both are completely different mathematical operations with no relationship.

14. What is a saddle point in neural network optimization?

A. A point where loss is always zero

B. A point that is a global minimum

C. A stationary point that is not necessarily a local minimum or maximum

D. A point where the learning rate becomes zero

15. Which statement best describes the Adam optimizer?

A. It uses only the current gradient

B. It combines momentum-like first-moment estimates with adaptive second-moment estimates

C. It never requires a learning rate

D. It works only with linear regression

Section B: Python Programming

Questions 16–30 · 15 marks

16. Which of the following is a valid Python list containing different data types?

A. (10, 3.5, True, 'AI')

B. {10, 3.5, True, 'AI'}

C. [10, 3.5, True, 'AI']

D. <10, 3.5, True, 'AI'>

17. What is the output of the following code?

```
nums = [10, 20, 30, 40, 50]

print(nums[-2])
```

A. 20

B. 30

C. 40

D. 50

18. What is the output?

```
nums = [0, 1, 2, 3, 4, 5]
print(nums[1:5:2])
```

A. [1, 2, 3, 4]

B. [1, 3]

C. [2, 4]

D. [1, 3, 5]

19. What does this code print?

```
nums = [1, 2]
nums.append([3, 4])
print(len(nums))
```

A. 2

B. 3

C. 4

D. Error

20. What is the output?

```
numbers = [10, 20, 30, 40, 50]
print(numbers[::-1])
```

A. [10, 20, 30, 40, 50]

B. [50, 40, 30, 20, 10]

C. [50, 40, 30, 20]

D. Error

21. What is the output?

```
items = [10, 20, 30]
```

```
items.insert(1, 99)
print(items)
```

A. [99, 10, 20, 30]

B. [10, 99, 20, 30]

C. [10, 20, 99, 30]

D. [10, 20, 30, 99]

22. Which statement about Python lists and tuples is correct?

A. Both are immutable

B. Lists are immutable, tuples are mutable

C. Lists are mutable, tuples are immutable

D. Both are unordered

23. What is the output?

```
nums = [10, 20, 30]
x = nums.pop()
print(x, nums)
```

A. 10 [20, 30]

B. 30 [10, 20]

C. 20 [10, 30]

D. None [10, 20]

24. What happens when discard(99) is called on a set that does not contain 99?

A. A KeyError is raised

B. The entire set is deleted

C. Nothing happens and no error is raised

D. 99 is inserted into the set

25. What is the output?

```
s = {1, 2, 2, 3, 3}

print(len(s))
```

A. 5

B. 4

C. 3

D. Error

26. Which syntax correctly creates a Python dictionary?

A. ['name': 'Alex', 'age': 20]

B. {'name': 'Alex', 'age': 20}

C. ('name': 'Alex', 'age': 20)

D. {'name', 'Alex', 'age', 20}

27. What is the main difference between a .py file and a .ipynb file?

A. .py supports Python 2 only

B. .ipynb is a notebook with executable cells and outputs, while .py is a Python source file

C. .ipynb cannot execute Python code

D. Both are compiled executables

28. What is printed?

```
x = 7
if x > 10:
        print('A')
elif x > 5:
        print('B')
else:
        print('C')
```

A. A

B. B

C. C

D. A and B

29. What is the output?

```
i = 1
while i < 4:
        print(i, end=' ')
        i += 1
```

A. 1 2 3

B. 1 2 3 4

C. 0 1 2 3

D. Infinite loop

30. What is the output?

```
def calculate(a, b=2):
    return a * b
print(calculate(5))
```

A. 2

B. 5

C. 10

D. Error

Ready to submit?

30 unanswered

Submit Assessment Answer all 30 questions to enable submission.

Assessment design: 1 mark per question, 30 marks total, no negative marking. Questions cover neural network fundamentals, activation functions, derivatives, optimization, Python data types, collections, CRUD operations, slicing, control flow, loops, and functions.